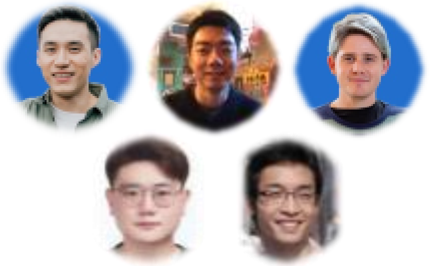
Sampling with Schrödinger Bridge

An Adjoint-Matching Perspective



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Research Scientist, FAIR, Meta

Outline

- Sampling problem setup
- Diffusion sampler
- Adjoint-based diffusion sampler
- Adjoint Schrodinger Bridge Sampler *(NeurIPS 2025 Oral)*

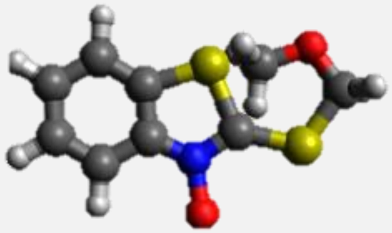
Sampling Problem

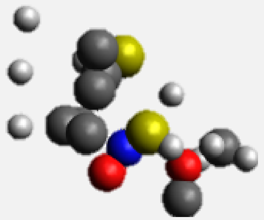
Boltzmann Distribution

$$\rho_{\text{target}}(x) := \frac{1}{Z}\exp(-E(x)), \quad Z := \int_{\mathcal{X}} \exp(-E(x))\mathrm{d}x$$

unnormalized energy function
(differentiable)

normalization constant
(usually intractable)

$E($  $)$ = 0.1

$E($  $)$ = 100

low energy → **stable** structure → **likely** to appear → **high** probability
high energy → **unstable** structure → **unlikely** to appear → **low** probability

Sampling Problem

Boltzmann Distribution

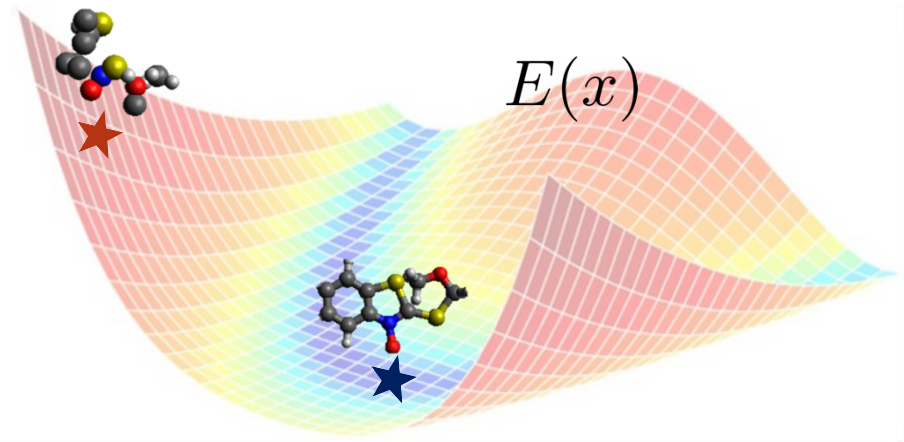
$$\rho_{\text{target}}(x) := \frac{1}{Z}\exp(-E(x)), \quad Z := \int_{\mathcal{X}} \exp(-E(x))\mathrm{d}x$$

unnormalized energy function
(differentiable)

normalization constant
(usually intractable)

Problem Statement

Given $E(x)$, generate $X \sim \rho_{\text{target}}$.



Sampling Method

Given $E(x)$, generate $X \sim \rho_{\text{target}}$, where $\rho_{\text{target}}(x) := \frac{1}{Z}\exp(-E(x))$

MCMC methods

Metropolis-Hastings
Annealed Importance Sampling
Hamiltonian Monte Carlo
...

Metropolis-Adjusted Langevin Algorithm

$$\mathrm{d}X_t = \underbrace{-\nabla E(X_t)}_{\text{exploitation}}\mathrm{d}t + \underbrace{\sqrt{2}\mathrm{d}W_t}_{\text{exploration}}$$

Learning-based methods

Diffusion samplers

$$\mathrm{d}X_t = [\underbrace{f_t(X_t)}_{\text{prescribed base drift}} + \underbrace{u_t^\theta(X_t)}_{\text{learned control drift}}]\mathrm{d}t + \underbrace{\sigma_t}_{\text{prescribed noise schedule}}\mathrm{d}W_t,$$

$$X_0 \sim \rho_{\text{source}}$$

- ✗ slow mixing rate
- ✗ expansive energy evaluation

Diffusion Sampler

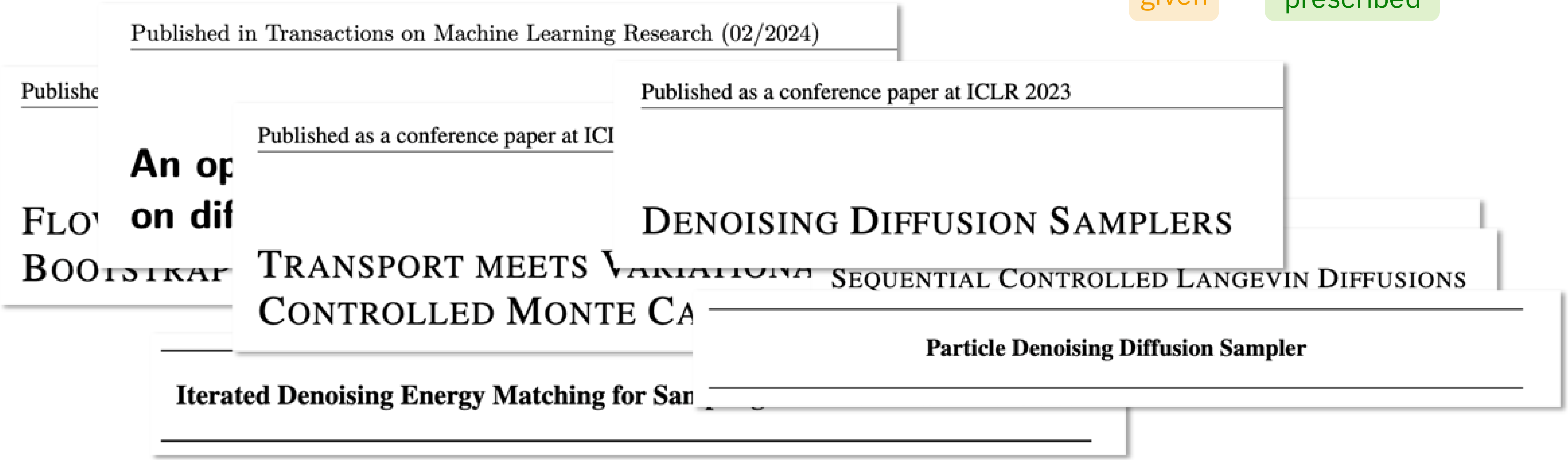
Find $u_t^\theta(\cdot)$ such that $X_1 \sim \rho_{\text{target}}$, where

$$X_0 \sim \rho_{\text{source}}$$

$$\mathrm{d}X_t = [f_t(X_t) + u_t^\theta(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t,$$

$$\rho_{\text{target}}(x) := \frac{1}{Z}\exp(-E(x))$$

given prescribed



Diffusion Sampler

Find $u_t^\theta(\cdot)$ such that $X_1 \sim \rho_{\text{target}}$, where $X_0 \sim \rho_{\text{source}}$

$$\mathrm{d}X_t = [f_t(X_t) + u_t^\theta(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t,$$

$$\rho_{\text{target}}(x) := \frac{1}{Z}\exp(-E(x))$$

given prescribed

Matching methods

$$\arg\min_\theta \mathbb{E}\left[\|u_t^\theta(X_t) - \square\|^2\right]$$

✓ scalable

Non-Matching methods

$$\nabla_\theta \mathcal{L}(X_1) \qquad \frac{\mathrm{d}p^u(X)}{\mathrm{d}q(X)}$$

✗ backpropagation through SDE, higher-order derivatives

Diffusion Sampler

Find $u_t^\theta(\cdot)$ such that $X_1 \sim \rho_{\text{target}}$, where $X_0 \sim \rho_{\text{source}}$
$$\mathrm{d}X_t = [f_t(X_t) + u_t^\theta(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t,$$
$$\rho_{\text{target}}(x) := \frac{1}{Z}\exp(-E(x))$$

given

prescribed

Matching methods

$$\arg\min_\theta \mathbb{E}\left[\|u_t^\theta(X_t) - \square\|^2\right]$$

score/flow/bridge matching

requires $X_1 \sim \rho_{\text{target}}$

adjoint matching

does NOT require $X_1 \sim \rho_{\text{target}}$

Adjoint Matching

Method for Stochastic Optimal Control (SOC)

$$\min_u \mathbb{E}\left[\int_0^1 \underbrace{\frac{1}{2\sigma_t^2}\|u_t(X_t)\|^2}_{\text{control cost / kinetic energy}} \mathrm{d}t + \underbrace{g(X_1)}_{\text{terminal cost / reward}}\right]$$

$$\text{s.t.}\ \ \mathrm{d}X_t = [f_t(X_t) + u_t(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t, \qquad X_0 \sim \rho_{\text{source}}$$

$\Updownarrow$

$$u^\star = \arg\min_u \mathbb{E}_{p^{\bar{u}}}\left[\|u_t(X_t) - \Phi_t(\boldsymbol{X}) \cdot \nabla g(X_1)\|^2\right]$$

$$\bar{u} = \mathtt{stopgrad}(u)$$

- ✓ Matching objective
- ✓ Does *not* require $X_1 \sim \rho_{\text{target}}$
- ✓ Detached model samples

Nam et al., *Adjoint Matching: Fine-tuning Flow and Diffusion Generative Models with Memoryless Stochastic Optimal Control,* ICLR 2025

Adjoint Matching

Method for Stochastic Optimal Control (SOC)

$$\min_u \mathbb{E}\left[\int_0^1 \frac{1}{2\sigma_t^2}\|u_t(X_t)\|^2 \mathrm{d}t + g(X_1)\right]$$

$$\text{s.t.}\ \ \mathrm{d}X_t = [f_t(X_t) + u_t(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad X_0 \sim \rho_{\text{source}}$$

$\Updownarrow$ $\Downarrow$

$$u^\star = \arg\min_u \mathbb{E}_{p^{\bar{u}}}\left[\|u_t(X_t) - \Phi_t(\boldsymbol{X}) \cdot \nabla g(X_1)\|^2\right]$$

$$\bar{u} = \mathtt{stopgrad}(u)$$

$$p^\star(X_1|X_0) \propto p^{\text{base}}(X_1|X_0) e^{-g(X_1)}$$

optimal distribution induced by u^*

base distribution

exponentially tilted

$$\mathrm{d}X_t = f_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad X_0 \sim \rho_{\text{source}}$$

Adjoint Matching

Application for sampling

$$\min_u \mathbb{E}\left[\int_0^1 \frac{1}{2\sigma_t^2}\|u_t(X_t)\|^2 \mathrm{d}t + g(X_1)\right]$$

$$\text{s.t.}\ \ \mathrm{d}X_t = [f_t(X_t) + u_t(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad X_0 \sim \rho_{\text{source}}$$

$\Updownarrow$ $\Downarrow$

$$u^\star = \arg\min_u \mathbb{E}_{p^{\bar{u}}}\left[\|u_t(X_t) - \Phi_t(\boldsymbol{X}) \cdot \nabla g(X_1)\|^2\right]$$

$$\bar{u} = \texttt{stopgrad}(u)$$

$$p^\star(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

Design g such that $\rho_{\text{target}}(X_1) = \int p^\star(X_1|X_0)\rho_{\text{source}}(X_0)\mathrm{d}X_0$

This is Schrödinger Bridge!

Connection to Schrödinger Bridge

SOC Optimal Solution

$$p^\star(X_0, X_1) = p^{\text{base}}(X_0, X_1) e^{-g(X_1) + V_0(X_0)}, \quad V_0(x) = -\log \int p^{\text{base}}_{1|0}(y|x) e^{-g(y)} \mathrm{d}y$$

$$\Updownarrow$$

SOC Optimal Solution (reparametrized)

$$p^\star(X_0, X_1) = p^{\text{base}}(X_0, X_1) \frac{\varphi_1(X_1)}{\varphi_0(X_0)}, \quad \varphi_0(x) = \int p^{\text{base}}_{1|0}(y|x) \varphi_1(y) \mathrm{d}y, \quad \varphi_1(\cdot) = e^{-g(\cdot)}$$

Design g such that $\rho_{\text{target}}(X_1) = \int p^\star(X_0, X_1) \mathrm{d}X_0$

$$\Leftrightarrow \rho_{\text{target}}(X_1) = \varphi_1(X_1) \int p^{\text{base}}(X_1|X_0) \frac{\rho_{\text{prior}}(X_0)}{\varphi_0(X_0)} \mathrm{d}X_0$$

$$\Leftrightarrow \frac{\rho_{\text{target}}(X_1)}{\varphi_1(X_1)} = \int p^{\text{base}}(X_1|X_0) \frac{\rho_{\text{prior}}(X_0)}{\varphi_0(X_0)} \mathrm{d}X_0$$

$$\Leftrightarrow \hat{\varphi}_1(X_1) = \int p^{\text{base}}(X_1|X_0) \hat{\varphi}_0(X_0) \mathrm{d}X_0$$

Connection to Schrödinger Bridge

SOC Optimal Solution

$$p^\star(X_0, X_1) = p^{\text{base}}(X_0, X_1) e^{-g(X_1)+V_0(X_0)}, \quad V_0(x) = -\log \int p^{\text{base}}_{1|0}(y|x) e^{-g(y)} \mathrm{d}y$$

Design g such that $\rho_{\text{target}}(X_1) = \int p^\star(X_0, X_1) \mathrm{d}X_0$

Conditions to Sample Target Distr.

Suppose $\varphi_1(\cdot) = e^{-g(\cdot)}$ satisfy

$$\varphi_0(x) = \int p^{\text{base}}_{1|0}(y|x) \varphi_1(y) \mathrm{d}y, \quad \varphi_0(x)\hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

$$\hat{\varphi}_1(y) = \int p^{\text{base}}_{1|0}(y|x) \hat{\varphi}_0(x) \mathrm{d}x, \quad \varphi_1(y)\hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

SB Theory: There's a *unique* g such that these equations hold.

(up to additive constant)

Adjoint-based Diffusion Sampler

Variational Inference
(Diffusion models → Stochastic Optimal Control)

Scalable Computational Method
(SOC → adjoint matching)

Probabilistic Characteristics

Adjoint Schrödinger Bridge Sampler

Variational Inference
(Diffusion models → Stochastic Optimal Control)

$\Updownarrow$ $\Downarrow$

Scalable Computational Method
(SOC → adjoint matching)

$\Rightarrow$

$$p^{\star}(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

$$g(X_1) := \log \hat{\varphi}_1(X_1) + E(X_1)$$

$$\varphi_0(x) = \int p^{\text{base}}_{1|0}(y|x)\varphi_1(y)\mathrm{d}y, \quad \varphi_0(x)\hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

$$\hat{\varphi}_1(y) = \int p^{\text{base}}_{1|0}(y|x)\hat{\varphi}_0(x)\mathrm{d}x, \quad \varphi_1(y)\hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

Adjoint Schrödinger Bridge Sampler

$$\min_u \mathbb{E}\left[\int_0^1 \frac{1}{2\sigma_t^2}\|u_t(X_t)\|^2 \mathrm{d}t + \log \hat{\varphi}_1(X_1) + E(X_1)\right]$$

$$\text{s.t. } \mathrm{d}X_t = u_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad X_0 \sim \rho_{\text{source}}$$

$\Updownarrow$ $\quad\quad\quad\quad$ $\Downarrow$

Scalable Computational Method
(SOC → adjoint matching)

$$\Longrightarrow \quad p^\star(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

$$g(X_1) := \log \hat{\varphi}_1(X_1) + E(X_1)$$

$$\varphi_0(x) = \int p_{1|0}^{\text{base}}(y|x)\varphi_1(y)\mathrm{d}y, \quad \varphi_0(x)\hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

$$\hat{\varphi}_1(y) = \int p_{1|0}^{\text{base}}(y|x)\hat{\varphi}_0(x)\mathrm{d}x, \quad \varphi_1(y)\hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

Adjoint Schrödinger Bridge Sampler

intractable

$$\min_u \mathbb{E}\left[\int_0^1 \frac{1}{2\sigma_t^2}\|u_t(X_t)\|^2 \mathrm{d}t + \log\hat{\varphi}_1(X_1) + E(X_1)\right]$$

$$\text{s.t. } \mathrm{d}X_t = u_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad X_0 \sim \rho_{\text{source}}$$

$\Updownarrow$ $\Downarrow$

$$u^\star = \arg\min_u \mathbb{E}_{p^{\bar{u}}}\left[\|u_t(X_t) + \sigma_t^2\left(\nabla E + \nabla \log\hat{\varphi}_1\right)(X_1)\|^2\right]$$

$$\nabla \log\hat{\varphi}_1 = \arg\min_h \mathbb{E}_{p^{\bar{u}}}\left[\|h(X_1) - \nabla_{x_1}\log p^{\text{base}}(X_1|X_0)\|^2\right]$$

$$\bar{u} = \texttt{stopgrad}(u)$$

$\Longrightarrow$

$$p^\star(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

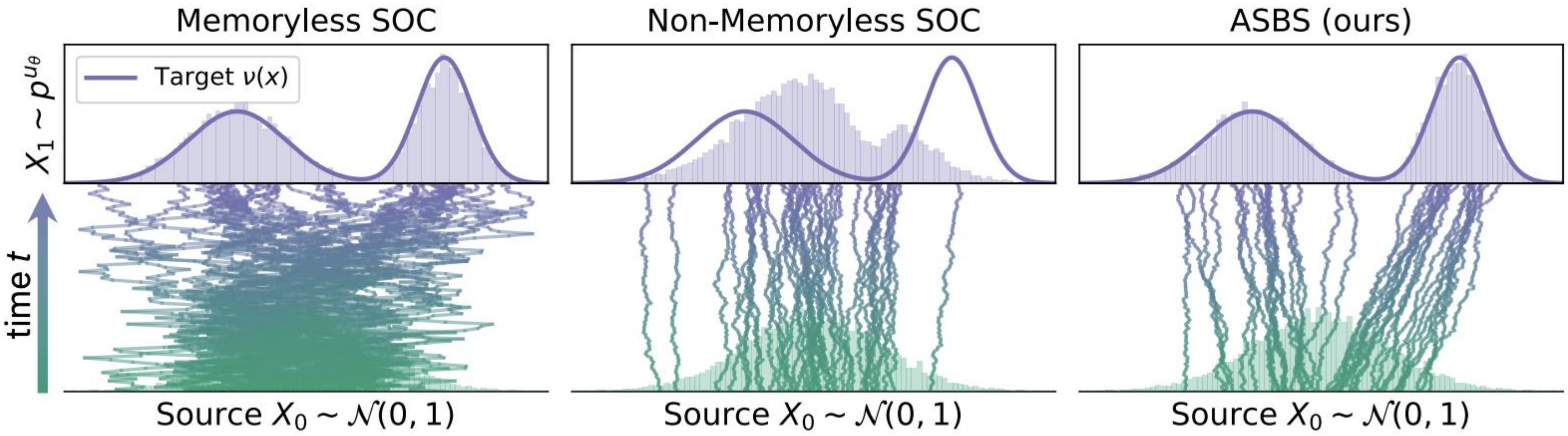
$$g(X_1) := \log\hat{\varphi}_1(X_1) + E(X_1)$$

$$\varphi_0(x) = \int p_{1|0}^{\text{base}}(y|x)\varphi_1(y)\mathrm{d}y, \quad \varphi_0(x)\hat{\varphi}_0(x) = \rho_{\text{prior}}(x),$$

$$\hat{\varphi}_1(y) = \int p_{1|0}^{\text{base}}(y|x)\hat{\varphi}_0(x)\mathrm{d}x, \quad \varphi_1(y)\hat{\varphi}_1(y) = \rho_{\text{target}}(y),$$

Adjoint Matching **Corrector Matching**

Adjoint Schrödinger Bridge Sampler



Adjoint Schrödinger Bridge Sampler

	MW-5 (d=5)	DW-4 ($d=8$)		LJ-13 ($d=39$)		LJ-55 ($d=165$)	
Method	Sinkhorn ↓	$\mathcal{W}_2$ ↓	$E(\cdot)$ $\mathcal{W}_2$ ↓	$\mathcal{W}_2$ ↓	$E(\cdot)$ $\mathcal{W}_2$ ↓	$\mathcal{W}_2$ ↓	$E(\cdot)$ $\mathcal{W}_2$ ↓
PDDS (Phillips et al., 2024)	—	$0.92_{\pm 0.08}$	$0.58_{\pm 0.25}$	$4.66_{\pm 0.87}$	$56.01_{\pm 10.80}$	—	—
SCLD (Chen et al., 2025)	$0.44_{\pm 0.06}$	$1.30_{\pm 0.64}$	$0.40_{\pm 0.19}$	$2.93_{\pm 0.19}$	$27.98_{\pm 1.26}$	—	—
PIS (Zhang and Chen, 2022)	$0.65_{\pm 0.25}$	$0.68_{\pm 0.28}$	$0.65_{\pm 0.25}$	$1.93_{\pm 0.07}$	$18.02_{\pm 1.12}$	$4.79_{\pm 0.45}$	$228.70_{\pm 131.27}$
DDS (Vargas et al., 2023)	$0.63_{\pm 0.24}$	$0.92_{\pm 0.11}$	$0.90_{\pm 0.37}$	$1.99_{\pm 0.13}$	$24.61_{\pm 8.99}$	$4.60_{\pm 0.09}$	$173.09_{\pm 18.01}$
LV-PIS (Richter and Berner, 2024)	—	$1.04_{\pm 0.29}$	$1.89_{\pm 0.89}$	—	—	—	—
iDEM (Akhound-Sadegh et al., 2024)	—	$0.70_{\pm 0.06}$	$0.55_{\pm 0.14}$	$1.61_{\pm 0.01}$	$30.78_{\pm 24.46}$	$4.69_{\pm 1.52}$	$93.53_{\pm 16.31}$
AS (Havens et al., 2025)	$0.32_{\pm 0.06}$	$0.62_{\pm 0.06}$	$0.55_{\pm 0.12}$	$1.67_{\pm 0.01}$	$2.40_{\pm 1.25}$	$4.50_{\pm 0.05}$	$58.04_{\pm 20.98}$
ASBS (**Ours**)	$0.15_{\pm 0.02}$	$0.38_{\pm 0.05}$	$0.19_{\pm 0.03}$	$1.59_{\pm 0.00}$	$1.28_{\pm 0.22}$	$4.00_{\pm 0.03}$	$27.69_{\pm 3.86}$

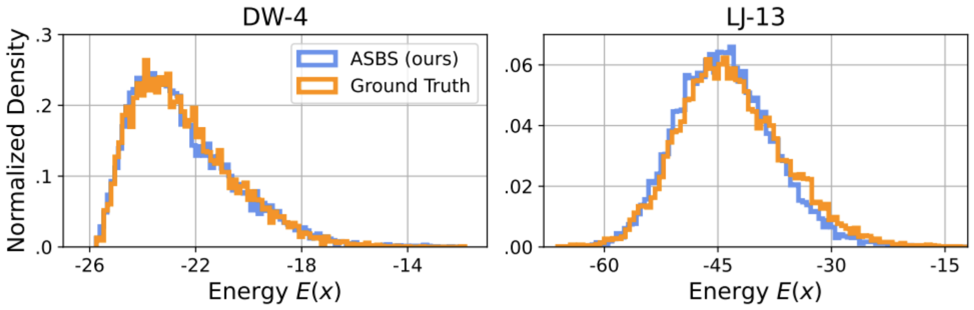


Figure 3 The energy histograms of DW-4 and LJ-13 from Table 2. ASBS generates samples whose energy profiles closely match those of the ground-truth samples.

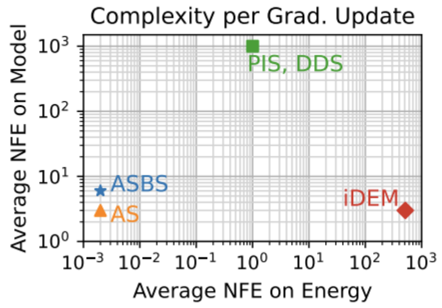


Figure 4 Complexity w.r.t. the number of function evaluation (NFE) on LJ-13 potential.

Adjoint Schrödinger Bridge Sampler

Large-Scale Amortized Conformer Generation

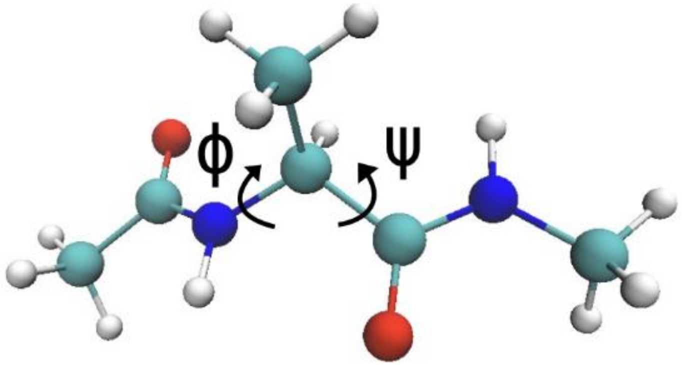
$$\rho_{\text{target}}(x|g) = e^{-\frac{1}{\tau}E(x|g)}, \quad g \in \mathcal{G}, \quad \tau \ll 1$$

graph structure conformer

	without relaxation				with relaxation			
	SPICE		GEOM-DRUGS		SPICE		GEOM-DRUGS	
Method	Coverage ↑	AMR ↓	Coverage ↑	AMR ↓	Coverage ↑	AMR ↓	Coverage ↑	AMR ↓
RDKit ETKDG (Riniker and Landrum, 2015)	$56.94_{\pm 35.82}$	$1.04_{\pm 0.52}$	$50.81_{\pm 34.69}$	$1.15_{\pm 0.61}$	$70.21_{\pm 31.70}$	$0.79_{\pm 0.44}$	$62.55_{\pm 31.67}$	$0.93_{\pm 0.53}$
AS (Havens et al., 2025)	$56.75_{\pm 38.15}$	$0.96_{\pm 0.26}$	$36.23_{\pm 33.42}$	$1.20_{\pm 0.43}$	$82.41_{\pm 25.85}$	$0.68_{\pm 0.28}$	$64.26_{\pm 34.57}$	$0.89_{\pm 0.45}$
ASBS w/ Gaussian prior (**Ours**)	$73.04_{\pm 31.95}$	$0.83_{\pm 0.24}$	$50.23_{\pm 35.98}$	$1.05_{\pm 0.43}$	$88.26_{\pm 20.57}$	$0.60_{\pm 0.24}$	$72.32_{\pm 29.68}$	$0.77_{\pm 0.35}$
ASBS w/ harmonic prior (**Ours**)	$74.05_{\pm 31.61}$	$0.82_{\pm 0.23}$	$53.14_{\pm 35.69}$	$1.03_{\pm 0.42}$	$88.71_{\pm 18.63}$	$0.59_{\pm 0.24}$	$72.77_{\pm 29.94}$	$0.78_{\pm 0.35}$

Adjoint Schrödinger Bridge Sampler

Alanine dipeptide



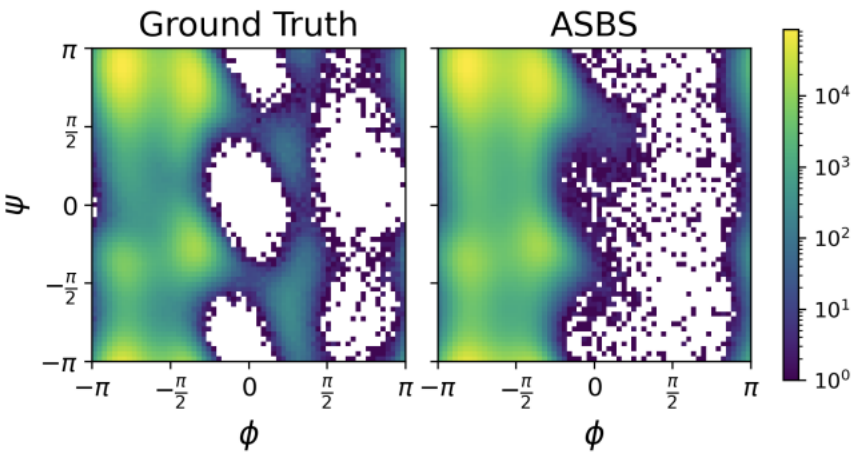


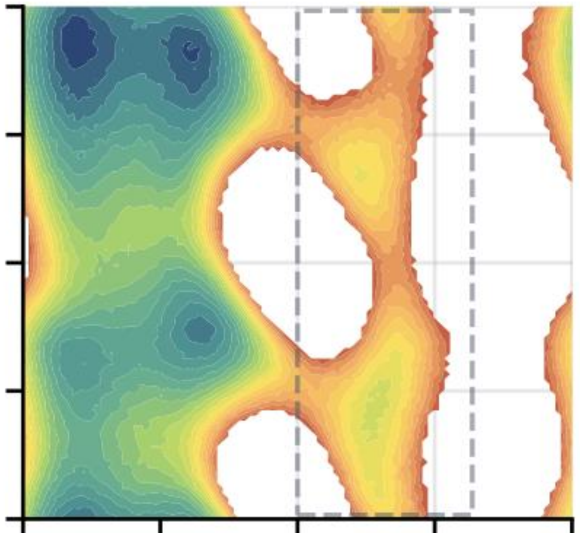
Table 3 Comparison between diffusion samplers on sampling the molecular Boltzmann distribution of the alanine dipeptide. We report the KL divergence D_{KL} for the 1D marginal across five torsion angles and the Wasserstein-2 $\mathcal{W}_2$ on jointly (ϕ, ψ), known as Ramachandran plots (see Figure 5). Best results are highlighted.

	D_{KL} on each torsion's marginal ↓					$\mathcal{W}_2$ on joint ↓
Method	ϕ	ψ	γ_1	γ_2	γ_3	(ϕ, ψ)
PIS (Zhang and Chen, 2022)	0.05±0.03	0.38±0.49	5.61±1.24	4.49±0.03	4.60±0.03	1.27±1.19
DDS (Vargas et al., 2023)	0.03±0.01	0.16±0.07	2.44±0.96	0.03±0.00	0.03±0.00	0.68±0.09
AS (Havens et al., 2025)	0.09±0.09	0.04±0.04	0.17±0.17	0.56±0.09	0.51±0.06	0.65±0.52
ASBS (**Ours**)	0.02±0.00	0.01±0.00	0.03±0.01	0.02±0.00	0.02±0.00	0.25±0.01

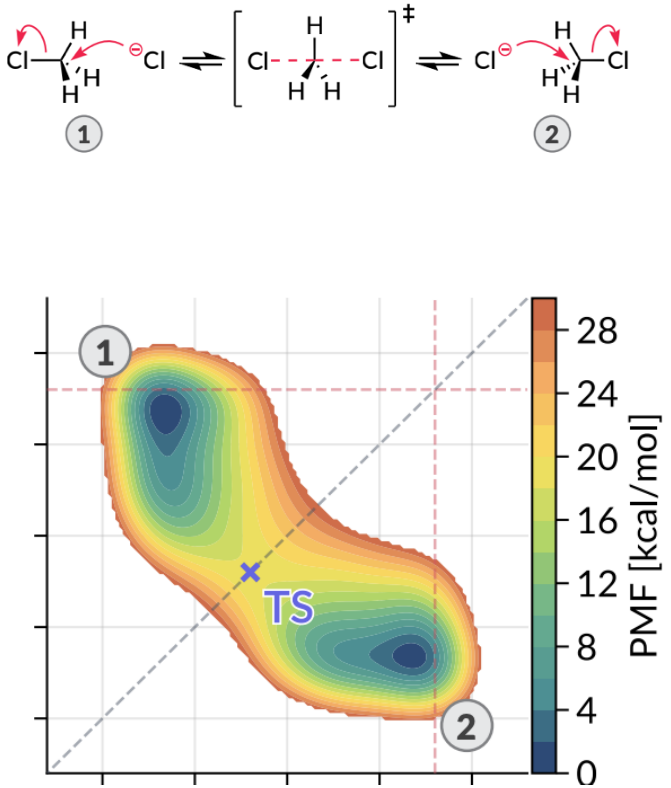
*image credit to FAB (Midgley et al 2022)

Adjoint Schrödinger Bridge Sampler

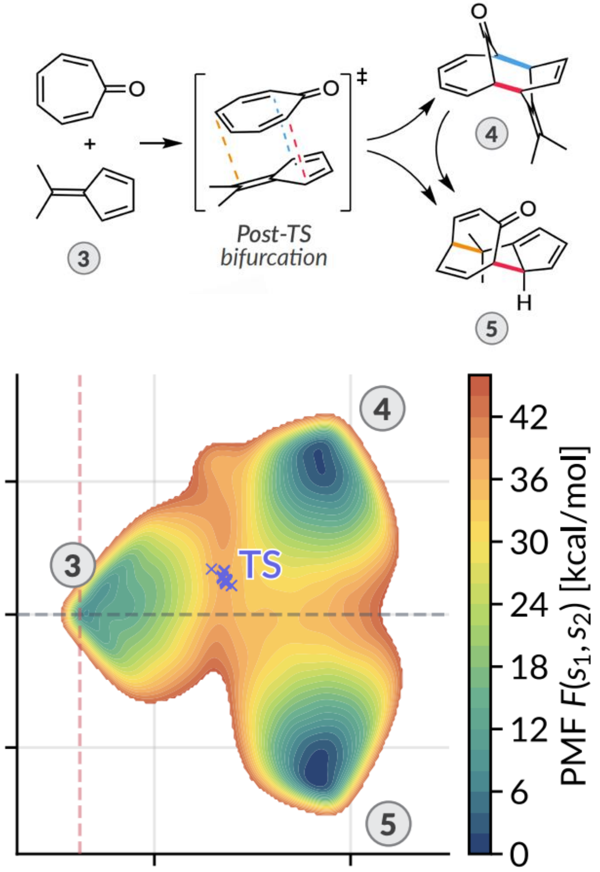
Adding chemical-based exploration



Alanine dipeptide



Reactive landscapes of Nucleophilic substitution (S_N2)

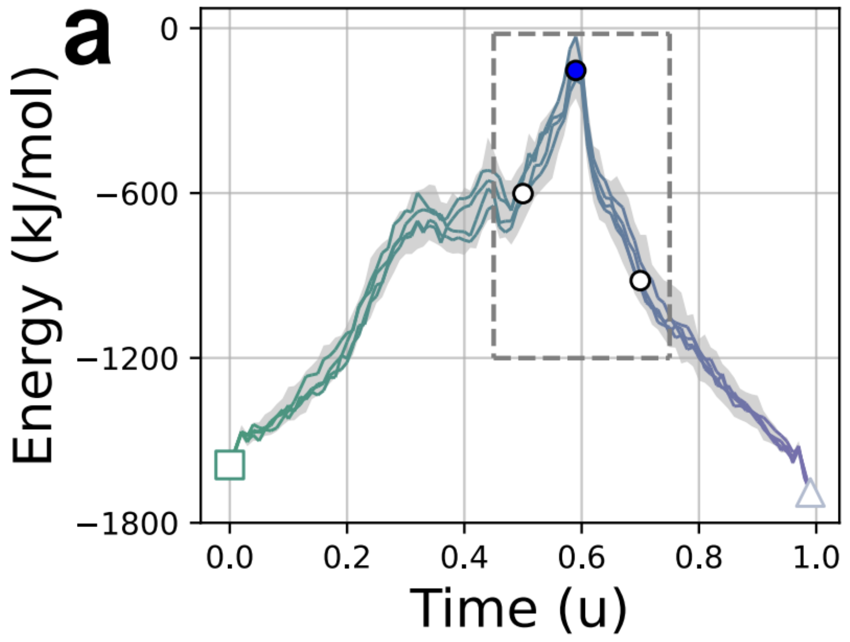


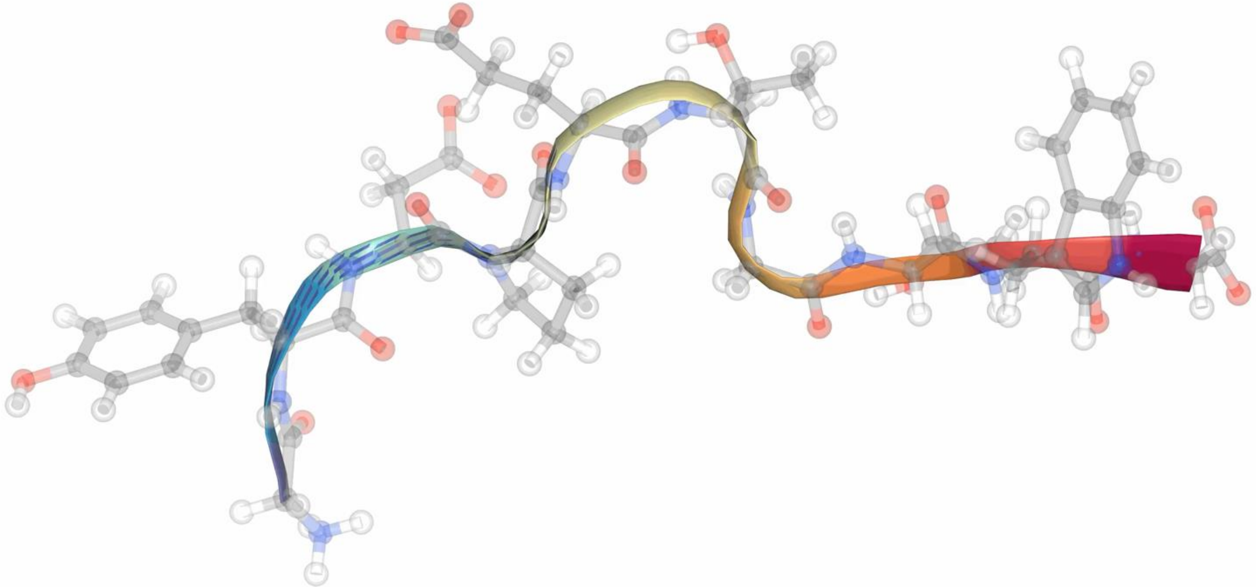
Reactive landscapes of Cycloaddition with post-TS bifurcation

Nam et al., *Enhancing Diffusion-Based Sampling with Molecular Collective Variables*, preprint 2025

Transition Path Sampling

Protein folding (Chignolin)





Park et al., *Functional Adjoint Sampler: Scalable Sampling on Infinite Dimensional Spaces*, preprint 2025

Adjoint-based Diffusion Sampler

Variational Inference / Stochastic Optimal Control

$$\min_u \mathbb{E}\left[\int_0^1 \frac{1}{2\sigma_t^2}\|u_t(X_t)\|^2 \mathrm{d}t + g(X_1)\right]$$

$$\text{s.t.}\ \ \mathrm{d}X_t = [f_t(X_t) + u_t(X_t)]\mathrm{d}t + \sigma_t \mathrm{d}W_t, \quad X_0 \sim \rho_{\text{source}}$$

$\Updownarrow$ $\qquad\qquad$ $\Downarrow$

Scalable Adjoint-based Method

$$u^\star = \arg\min_u \mathbb{E}_{p^{\bar{u}}}\left[\|u_t(X_t) - \Phi_t(\boldsymbol{X}) \cdot \nabla g(X_1)\|^2\right]$$

$$\bar{u} = \texttt{stopgrad}(u)$$

$\Longrightarrow$

Probabilistic Characteristics

$$p^\star(X_1|X_0) \propto p^{\text{base}}(X_1|X_0)e^{-g(X_1)}$$

[1] *Adjoint Sampling*, ICML 2025.

[2] *Adjoint Schrödinger Bridge Sampler*, NeurIPS 2025 (*Oral*).

[3] *Non-equilibrium Annealed Adjoint Sampler*, NeurIPS 2025.

[4] *Well-Tempered Adjoint Schrödinger Bridge Sampler*, preprint 2025.

[5] *Functional Adjoint Sampler*, preprint 2025.

arXiv *code* *slides*